3

Solution: Its easy to see that Row 1 Column 1, Row 2 Column 3 and Row 3 Column 4 are pivots. ∴ Columns 1, 2 and 4 are pivot columns.

By Theorem 13, the bases for Col A are Columns 1, 2, 4, i.e.

$$\left\{\begin{bmatrix}1\\-3\\2\\-4\end{bmatrix}, \begin{bmatrix}-4\\5\\-3\\7\end{bmatrix}, \begin{bmatrix}2\\-1\\4\\2\end{bmatrix}\right\} \quad \text{Dimension: 3}$$

Since Column 2 has no pivots, it is the basis for Nul A.

$$\left\{\begin{bmatrix}-3\\9\\-6\\12\end{bmatrix}\right\} \rightsquigarrow \text{Wrong.}$$

$$\left[\begin{array}{cccc|c}1&-3&2&-4&0\\0&0&5&-7&0\\0&0&0&5&0\\0&0&0&0&0\end{array}\right]$$

1. $x_1 - 3x_2 + 2x_3 - 4x_4 = 0$
2. $5x_3 - 7x_4 = 0$
3. $5x_4 = 0 \Rightarrow x_4 = 0$

$5x_3 - 7(0) = 0 \Rightarrow x_3 = 0.$

$x_1 - 3x_2 + 2(0) - 4(0) = 0 \Rightarrow x_1 = 3x_2.$

$$x = \begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} = x_2\begin{bmatrix}3\\1\\0\\0\end{bmatrix} \rightsquigarrow \text{The basis of Nul A.}$$

Dimension: 1

11. $$A = \begin{bmatrix}1&2&-5&0&-1\\2&5&-8&4&3\\-3&-9&9&-7&-2\\3&10&-7&11&7\end{bmatrix} \sim \begin{bmatrix}\boxed{1}&2&-5&0&-1\\0&\boxed{1}&2&4&5\\0&0&0&\boxed{1}&2\\0&0&0&0&0\end{bmatrix}$$

Solution: Its easy to see that Row 1 Column 1, Row 2 Column 2 and Row 3 Column 4 are pivots. ∴ Columns 1, 2 and 4 are pivot columns.

By Theorem 13, the bases for Col A are Columns 1, 2 and 4, i.e.

$$\left\{\begin{bmatrix}1\\2\\-3\\3\end{bmatrix}, \begin{bmatrix}2\\5\\-9\\10\end{bmatrix}, \begin{bmatrix}0\\4\\-7\\11\end{bmatrix}\right\} \quad \text{Dimension: 3}$$

$$\begin{bmatrix}1&2&-5&0&-1&0\\0&1&2&4&5&0\\0&0&0&1&2&0\\0&0&0&0&0&0\end{bmatrix}$$

1. $x_1 + 2x_2 - 5x_3 - x_5 = 0$
2. $x_2 + 2x_3 + 4x_4 + 5x_5 = 0$
3. $x_4 + 2x_5 = 0$

$x_4 = -2x_5$

$x_2 + 2x_3 + 4(-2x_5) + 5x_5 = 0$

$\Rightarrow x_2 = -2x_3 + 3x_5$